Toronto education & income

Final revised report | Verified tables, charts and plain-English findings

Final revision date: October 20, 2026

Executive verdict

The descriptive approach is appropriate. All 143 numeric entries in the three results tables in the attached analytical review agree with the supplied SPSS data and independently recalculated differences. No numerical changes were needed. This final version corrects the audit count and references, removes draft instructions, states the population consistently and replaces all four charts with plots regenerated from the verified source data.

Review area	Verdict
Statistical approach	Appropriate descriptive comparisons; no inferential tests are shown.
Table verification	143 of 143 numeric entries in the three results tables passed independent checks.
Filtering / record counts	32 all-gender and 64 gender-specific cells before excluding education totals; 22 and 44 after exclusion. No missing selected medians.
Important correction	Work activity refers to 2020 for BOTH income years. This definition is applied consistently throughout the final report.
Causal or significance claims	Not supported. These are unadjusted group differences, not returns caused by education or effects caused by the pandemic.

What a manager should know

- University qualifications generally corresponded to higher median annual employment income, but credentials did not form a simple income ladder.
- Bachelor's graduates had median income of CAD 74,500 in 2019 and CAD 81,000 in 2020. These were 41.1% and 46.7% above high-school graduates, respectively.
- Women+ had lower median income than Men+ in every detailed education group in both years. More education did not consistently correspond to a smaller gap.
- The percentage gender gap narrowed in all 11 groups. Apprenticeship holders still had the largest gap in 2020 (40.1%); doctorate holders had the smallest (9.6%).
- Median income increased in every education group: from 1.7% for people with no credential to 8.7% for bachelor's graduates. These figures describe the selected population, not all Toronto workers.

Reading guide: "median" means the middle income, not the average. All monetary figures are Canadian dollars. "Toronto" refers to the Toronto census metropolitan area, not just the City of Toronto.

1. Are the methods appropriate?

Yes—for the descriptive questions actually answered. Select Cases, Frequencies, Sort Cases and Chart Builder are data-management or display procedures, not statistical hypothesis tests. Reading published group medians and calculating percentage differences is suitable here. There is no need to add p-values simply to make the report look more statistical.

Question	Appropriate analysis	What it does not establish
Income by education	Compare published medians and premiums relative to high school.	A causal financial return from obtaining a credential.
Gender income gap	Compare Men+ and Women+ medians within education; calculate the relative gap.	Equal pay for equal work, discrimination, or occupation-adjusted differences.
2019 versus 2020	Compare income-year medians in the same census-defined categories.	Pandemic causality, individual income trajectories, or the whole labour market.

Why ordinary individual-level tests would be misleading

Each SPSS row is an aggregate demographic cell, not a person. Ordinary t-tests, ANOVA, rank tests or individual-income regressions on these rows would not answer the stated person-level questions: they would treat published summaries as observations and ignore unequal group sizes, overlap, sampling design and uncertainty. A cell-level model is possible for a different, explicitly aggregate question, but is not a substitute for individual-level inference. Published population counts must not be used to duplicate medians into artificial person records.

For formal individual-level comparisons or adjusted gender/education relationships, use suitable microdata and survey weights/design information. For uncertainty in published medians, obtain suitable official standard errors or replicate-weight estimates. Neither is supplied here. A chi-square test could address a different question about categorical counts, not differences in median income.

Essential population correction [1–2]

Statistics Canada's table metadata explicitly defines work activity using weeks and usual full-time/part-time status in 2020. Full year means 49 weeks or more; full time means mostly 30 hours or more per week. Age and highest qualification are census characteristics, not separately observed 2019 characteristics. The identical total-population counts across income years in the SAV are consistent with this fixed classification.

Recommended scope statement: "2019 and 2020 median employment income among employment-income recipients in the Toronto CMA, aged 25–64 at the 2021 Census, who were classified as working full year, full time in 2020, grouped by highest qualification at the census."

This selection excludes people who did not meet the full-year, full-time work criterion in 2020, including people whose employment disruption took them outside that criterion. It therefore cannot measure the pandemic's overall effect on employment or income. Medians use recipients with employment income, positive or negative; recipient counts differ between years, so changes in group medians are not median individual pay rises. [1–2]

2. Verification findings and completed revisions

Check	Result
SPSS structure	147,840 aggregate records; 15 variables; Toronto CMA only; 2019 and 2020 income years; 2021 reference date.
Duplicate demographic keys	0 duplicates across the eight dimension variables.
All-gender selected cells	32 cells before excluding five overlapping education totals; 22 after exclusion.
Men+ / Women+ selected cells	64 before excluding education totals; 44 after exclusion (22 per year).
Missing medians	0 in the selected 96 cells, including education aggregates.
Table 1: education medians and changes	44 / 44 numeric entries agree with SAV/recalculation.
Table 2: gender medians and gaps	66 / 66 entries agree for 2019 and 2020.
Table 3: gender-gap changes	33 / 33 entries agree; changes use unrounded gaps.
Overall	143 / 143 numeric entries verified; no arithmetic corrections required.

Formula checks

Education premium (%) = (education median − high-school median) ÷ high-school median × 100. Gender gap (%) = (Men+ median − Women+ median) ÷ Men+ median × 100. Income change (%) = (2020 median − 2019 median) ÷ 2019 median × 100. Gap change (percentage points) = unrounded 2020 gap − unrounded 2019 gap.

Bachelor's premium increased from 41.0985% to 46.7391%: an increase of 5.6406 percentage points, correctly reported as 5.6 points. Master's premiums (61.0%, 66.7%) and doctorate premiums (93.2%, 99.3%) are also correct.

Revisions completed in this final report

- Retained the correct numerical results and consistently stated that work status is measured in 2020 for both income years.
- Corrected the audit description: the attached analytical review has three results tables containing 143 numeric entries. Summary and methods tables are not counted as results tables. This is a clean document with no pending tracked revisions.
- Used "aggregate cells," not respondent sample sizes: 22 all-gender and 44 gender-specific cells across both years after excluding five overlapping education aggregates.
- Retained Men+ / Women+ labels and explained that each includes some non-binary people distributed to protect confidentiality. [1–2]
- Described the outcome as annual employment income, which includes wages, salaries, commissions and net self-employment income—not salary alone or hourly pay. [1–2]
- Distinguished percentage gaps from dollar gaps: for no credential, the dollar gap remained CAD 10,800, and for bachelor's graduates it remained CAD 14,000.
- Retained CAD values as supplied and avoided claims about real purchasing-power growth, because the supplied SAV metadata does not establish the price basis.

Verification boundary: the attached analytical review's results tables were checked cell by cell against the supplied SAV; all four final charts were regenerated from those same verified values. Statistics Canada metadata was rechecked for the population and variable definitions. This does not independently authenticate every SAV value against a fresh source-cube extraction or certify an unprovided SPSS execution log.

3. Income by education and year-to-year change

Table 1. Median employment income and change by highest qualification. All-gender medians; CAD. Qualification labels are shortened as explained below; no external workbook is needed to interpret this report.

Highest qualification	2019	2020	CAD change	% change
No credential	46,000	46,800	+800	+1.7%
High school	52,800	55,200	+2,400	+4.5%
Apprenticeship	73,000	74,500	+1,500	+2.1%
Other trades certificate	54,000	56,800	+2,800	+5.2%
College / CEGEP	60,800	64,000	+3,200	+5.3%
University certificate below bachelor's	59,200	63,600	+4,400	+7.4%
Bachelor's degree	74,500	81,000	+6,500	+8.7%
University certificate above bachelor's	79,500	86,000	+6,500	+8.2%
Medicine / dentistry / veterinary / optometry	80,000	84,000	+4,000	+5.0%
Master's degree	85,000	92,000	+7,000	+8.2%
Earned doctorate	102,000	110,000	+8,000	+7.8%

Plain-English interpretation

Bachelor's graduates had median income CAD 21,700 above high-school graduates in 2019 and CAD 25,800 above them in 2020. In relative terms, the difference widened from 41.1% to 46.7%—an increase of 5.6 percentage points.

Doctorate holders had the highest median income among the 11 detailed groups in both years: CAD 102,000 and CAD 110,000. People with no credential had the lowest: CAD 46,000 and CAD 46,800.

However, not every additional type of qualification corresponded to higher income. Apprenticeship holders had higher medians than college graduates: CAD 73,000 versus CAD 60,800 in 2019, and CAD 74,500 versus CAD 64,000 in 2020. These are different educational pathways, not evenly spaced steps on one ladder.

Median incomes increased in all 11 groups. Bachelor's graduates had the largest percentage increase (+8.7%); doctorate holders had the largest dollar increase (+CAD 8,000). These are changes in group medians, not measured pay increases for each person.

Population for all results: Toronto CMA, census age 25–64, full-year/full-time work status in 2020, with employment income in the indicated year. No adjustment for occupation, experience, field of study or other group differences. Label key: "No credential" means no certificate, diploma or degree; "High school" includes equivalency; "Other trades certificate" means non-apprenticeship trades certificate or diploma; college and university certificate groups include diplomas. The professional-degree category includes medicine, dentistry, veterinary medicine and optometry. All categories refer to highest qualification.

4. Gender medians and percentage gaps

Table 2. Median employment income by gender and highest qualification, 2019 and 2020. CAD; percentage gap is the Women+ median shortfall relative to the Men+ median. Population and shortened qualification labels are defined in Section 3.

Qualification	Men+ 2019	Women+ 2019	Gap 2019	Men+ 2020	Women+ 2020	Gap 2020
No credential	50,400	39,600	21.4%	51,200	40,400	21.1%
High school	56,000	48,800	12.9%	58,400	51,600	11.6%
Apprenticeship	77,500	44,400	42.7%	79,500	47,600	40.1%
Other trades certificate	59,200	46,000	22.3%	61,200	49,600	19.0%
College / CEGEP	68,500	54,800	20.0%	71,500	58,000	18.9%
University certificate below bachelor's	66,000	54,800	17.0%	69,500	58,800	15.4%
Bachelor's degree	82,000	68,000	17.1%	88,000	74,000	15.9%
University certificate above bachelor's	85,000	75,000	11.8%	91,000	81,000	11.0%
Medicine / dentistry / veterinary / optometry	88,000	74,500	15.3%	90,000	78,000	13.3%
Master's degree	93,000	76,500	17.7%	100,000	84,000	16.0%
Earned doctorate	108,000	94,000	13.0%	115,000	104,000	9.6%

Plain-English interpretation

Women+ had lower median income than Men+ in every education group in both years. Higher qualifications did not consistently correspond to smaller gaps. For example, the 2019 bachelor's gap was 17.1%, compared with 12.9% for high-school graduates. This is a comparison of groups, not proof that completing a bachelor's degree increased any person's gender-related disadvantage.

Apprenticeship holders had the largest gap in both years: 42.7% in 2019 and 40.1% in 2020. In 2020, Men+ in this group had median income CAD 79,500 and Women+ CAD 47,600—a difference of CAD 31,900.

The smallest gap was among university certificate holders above bachelor level in 2019 (11.8%) and doctorate holders in 2020 (9.6%). For doctorate holders in 2020, the medians were CAD 115,000 for Men+ and CAD 104,000 for Women+, a difference of CAD 11,000.

A 40.1% gap means the Women+ group median was 40.1% below the Men+ group median. It does not mean every woman earned 40.1% less than a comparable man, or that men earned only 40.1% more than women; the denominator matters.

Men+ and Women+ are Statistics Canada's published categories, each including some non-binary persons. These figures are annual employment-income gaps, not hourly wage comparisons or equal-pay-for-equal-work measures. [1–2]

5. Did the gender gaps narrow?

Table 3. Gender gaps and change, by highest qualification. Percentage-point changes use unrounded values; negative values indicate narrowing. The same 2020 work-status selection applies to both income years.

Highest qualification	2019 gap	2020 gap	Change (pp)
No credential	21.4%	21.1%	-0.3
High school	12.9%	11.6%	-1.2
Apprenticeship	42.7%	40.1%	-2.6
Other trades certificate	22.3%	19.0%	-3.3
College / CEGEP	20.0%	18.9%	-1.1
University certificate below bachelor's	17.0%	15.4%	-1.6
Bachelor's degree	17.1%	15.9%	-1.2
University certificate above bachelor's	11.8%	11.0%	-0.8
Medicine / dentistry / veterinary / optometry	15.3%	13.3%	-2.0
Master's degree	17.7%	16.0%	-1.7
Earned doctorate	13.0%	9.6%	-3.4

Plain-English interpretation

Yes, the percentage gap narrowed in all 11 education groups. The largest narrowing was among doctorate holders (3.4 percentage points), followed by other trades certificate holders (3.3 points). The smallest narrowing was among people with no credential (0.3 points).

For bachelor's graduates, the gap narrowed from 17.1% to 15.9%, a decrease of 1.2 percentage points. Their absolute median-income gap stayed at CAD 14,000: CAD 82,000 minus CAD 68,000 in 2019, and CAD 88,000 minus CAD 74,000 in 2020. Both percentage and dollar comparisons are useful; they answer different questions.

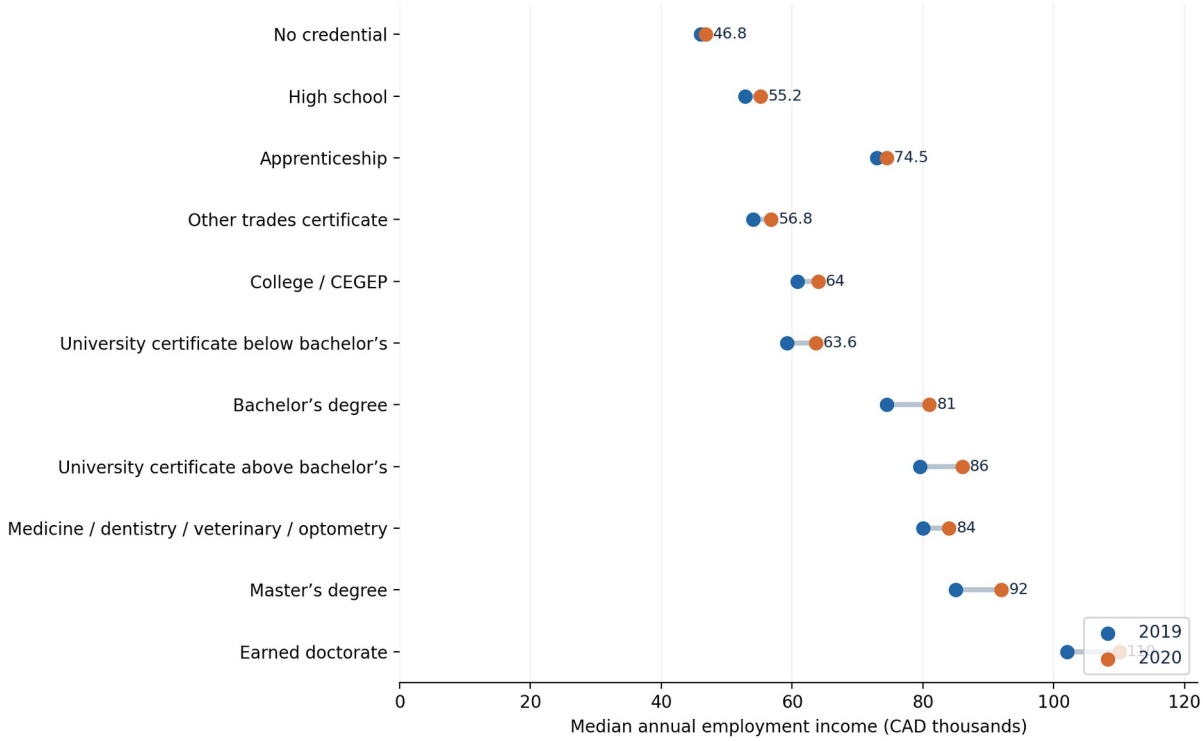
Changes are calculated before rounding. For example, the high-school gaps display as 12.9% and 11.6%, but the correct change is −1.2 percentage points, not −1.3, because the unrounded values are 12.8571% and 11.6438%.

Do not average the 11 gaps to estimate the overall Toronto gender gap. Each category represents a different population size, and a median for a combined population is not an average of its subgroup medians. For an overall result, use the published education-total median for each gender.

This comparison does not establish why the gaps narrowed. Occupation, industry, experience, actual annual hours, group composition and other factors are not adjusted for here. No statistical significance claim is made.

6. Visual summary: education and income

Income was higher in 2020 at every education level



Toronto CMA • Census age 25–64 • Full-year, full-time work status in 2020 for both income years
Employment-income recipients; group medians. Source: supplied SPSS dataset; Statistics Canada Table 98-10-0410-01.

How to read this chart

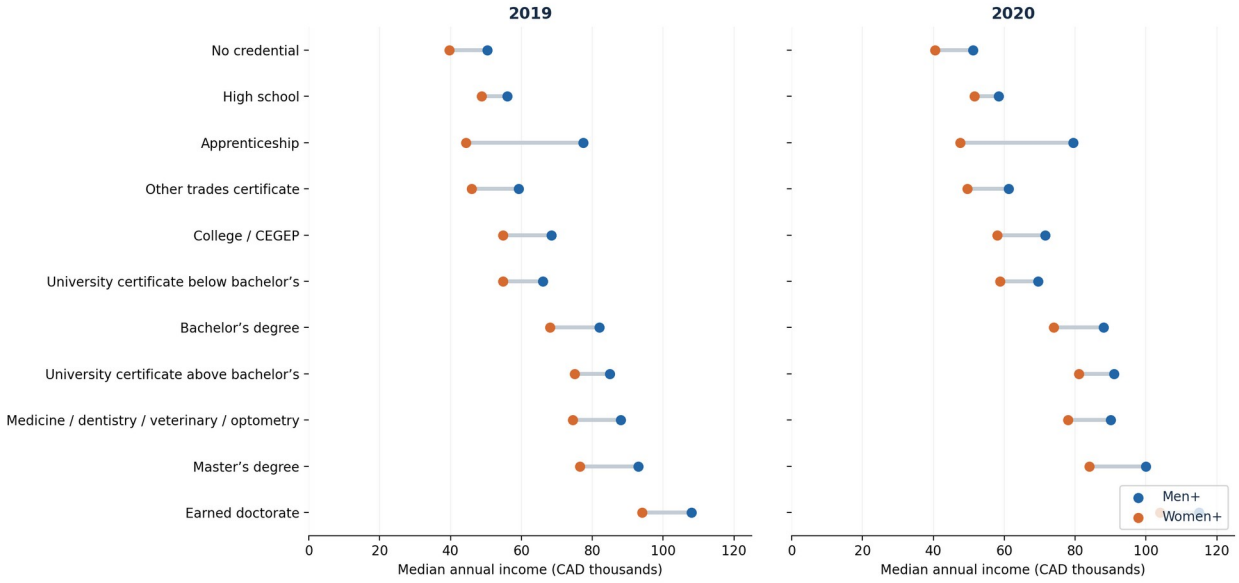
Two dots per row show the 2019 and 2020 medians. The connecting segment makes the change easy to see without suggesting that education categories are evenly spaced.

Client takeaway

All 2020 points lie to the right of the 2019 points: every group median increased. Doctorate holders remain highest; no-credential holders remain lowest. Apprenticeship medians exceed college medians in both years.

7. Visual summary: gender-specific income

Women+ had lower median income in every education group



Toronto CMA • Census age 25–64 • Full-year, full-time work status in 2020 for both income years
Employment-income recipients; group medians. Source: supplied SPSS dataset; Statistics Canada Table 98-10-0410-01.

How to read this chart

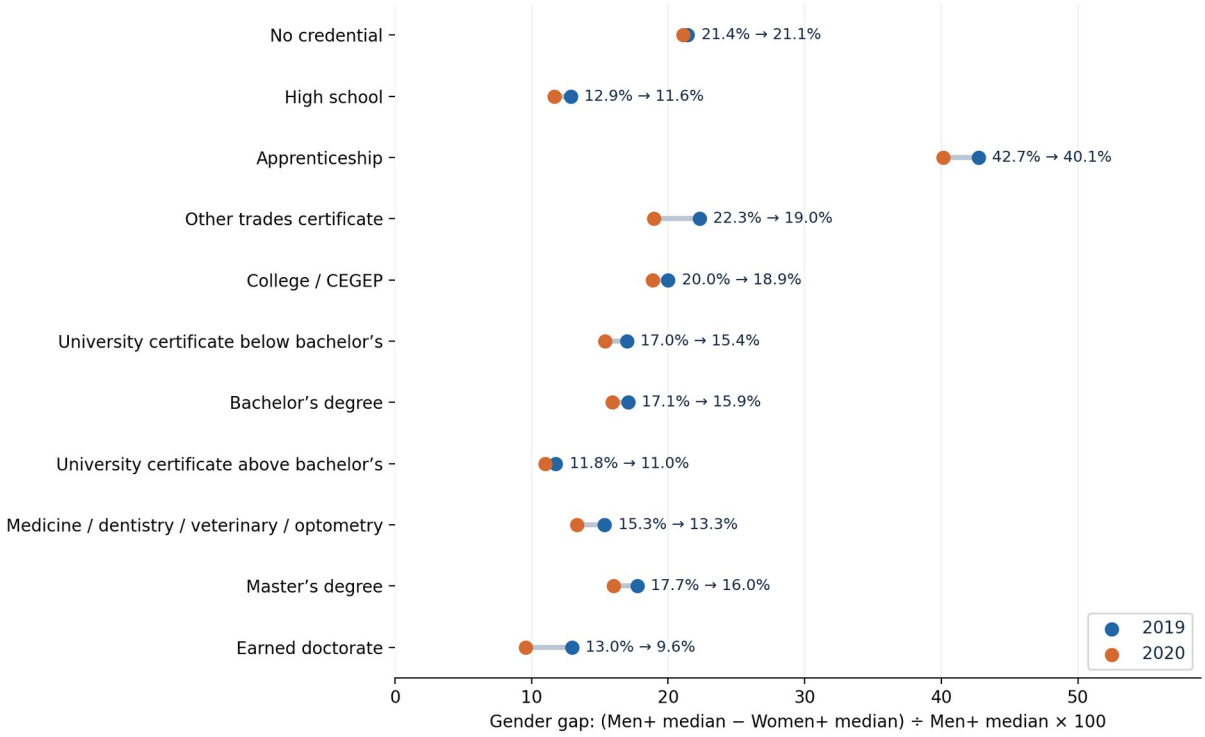
Separate panels use the same income scale and education order. Orange dots represent Women+ and blue dots Men+; the distance between them is the dollar difference between group medians.

Client takeaway

Women+ medians are below Men+ medians in all education groups and both years. The chart shows differences in annual employment income, not comparable-job hourly wages.

8. Visual summary: gender gap

Gender gaps narrowed, but remained uneven across credentials



Toronto CMA • Census age 25–64 • Full-year, full-time work status in 2020 for both income years
Employment-income recipients; group medians. Source: supplied SPSS dataset; Statistics Canada Table 98-10-0410-01.

How to read this chart

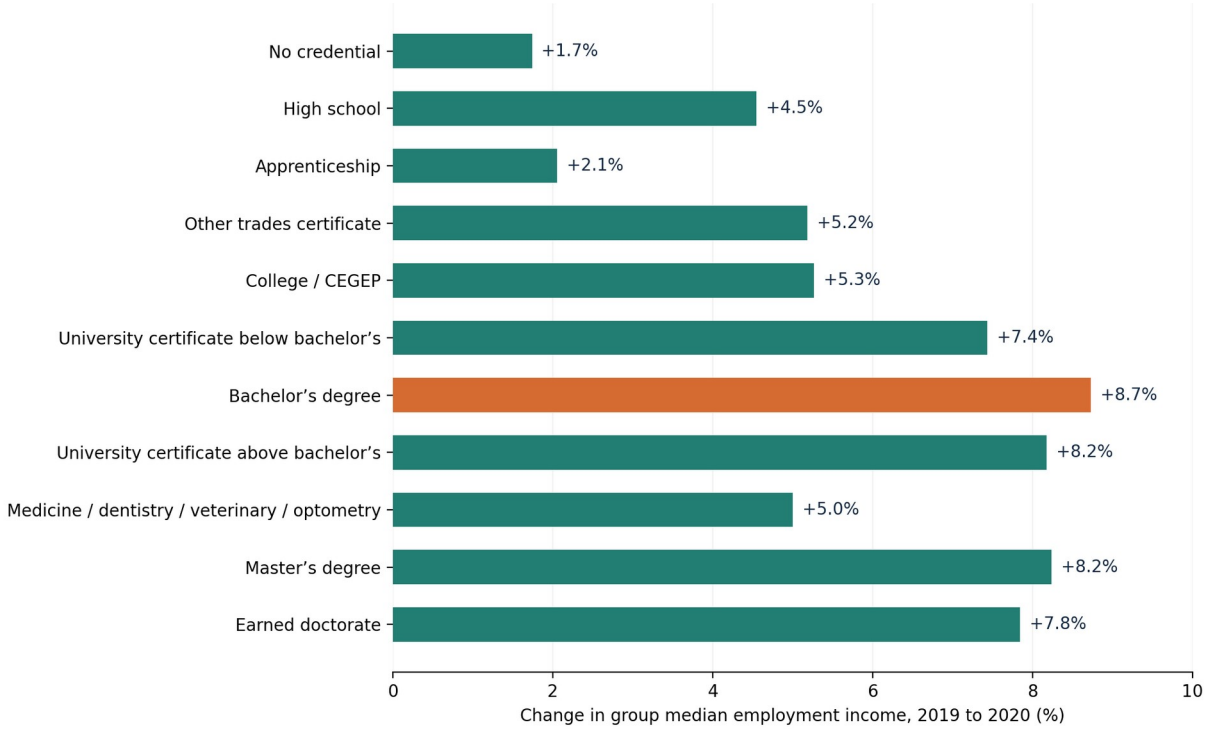
Paired dots show the percentage gap in each year; labels display 2019 → 2020. A shift to the left indicates a smaller percentage gap.

Client takeaway

Every gap moved left in 2020, but the pattern across qualifications remains uneven. Apprenticeship still has by far the largest gap. A higher credential does not reliably correspond to a smaller gap.

9. Visual summary: income growth

Bachelor’s graduates had the largest percentage increase



Toronto CMA • Census age 25–64 • Full-year, full-time work status in 2020 for both income years
Employment-income recipients; group medians. Source: supplied SPSS dataset; Statistics Canada Table 98-10-0410-01.

How to read this chart

Horizontal bars begin at zero and show percentage growth in the group median. The education order is held consistent across figures; the orange bar highlights bachelor’s graduates.

Client takeaway

Bachelor’s graduates had the strongest percentage growth (+8.7%). The pattern is not strictly increasing with education; apprenticeship grew only 2.1%, while the below-bachelor university certificate group grew 7.4%.

10. Limitations, sources and final recommendation

- Descriptive, unadjusted results: no proof of education causing income differences, of discrimination causing the gender gaps, or of COVID-19 causing the year-to-year changes.
- A 2020 work-status selection is applied to both income years. Results omit people outside that selected work pattern and cannot represent all workers or all people affected by job loss.
- The table is based on 2021 Census 25% sample data; published counts are population estimates, not raw respondent counts. No standard errors or replicate weights were supplied. [1]
- Census age, highest education and geography are census characteristics. Do not assume the person held the same credential, age category or location throughout both income years.
- The data contain group medians, means and counts, not individual income distributions. They cannot reveal within-group inequality or reconstruct valid individual-level records.
- There are no missing medians in the selected cells, but this does not establish that the original data are error-free or free of nonresponse, rounding and coverage limitations.
- Dollar values are retained as supplied. Inflation adjustment has not been independently established from the supplied file. No claim about changes in purchasing power is justified here.
- Verification covers all 143 numeric entries in the three results tables in the attached analytical review and the data underlying the four regenerated final charts. It does not authenticate the original extraction from Statistics Canada or certify an unprovided SPSS execution log.

Client-ready conclusion

Among Toronto CMA employment-income recipients aged 25–64 at the census and classified as full-year, full-time workers in 2020, higher qualifications generally corresponded to higher median employment income. Bachelor's graduates had a 41.1% median-income advantage over high-school graduates in 2019 and a 46.7% advantage in 2020. Women+ had lower medians than Men+ in every education group. Percentage gaps narrowed in all groups, but higher education did not consistently correspond to a smaller gap. These findings describe differences between groups; they do not establish their causes.

Sources and audit trail

Primary analytical sources: the user-supplied SPSS dataset and the attached analytical review. The 44 entries in Table 1, 66 in Table 2 and 33 in Table 3 were checked against source medians and recalculated quantities (143 checks in total). All passed. All four charts were rebuilt from these data. The 2020 work-activity definition was confirmed against the Statistics Canada metadata on October 20, 2026.

[1] Statistics Canada. Table 98-10-0410-01, Employment income statistics by highest level of education and major field of study (summary). Released October 4, 2023. Source-table information retained from the October 19, 2026 review; definitions reconfirmed through the metadata service on October 20, 2026. https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=9810041001

[2] Statistics Canada Web Data Service, getCubeMetadata, productId 98100410. Footnotes 2–4 (gender categories), 13–17 (2020 work activity, weeks and hours), 24 (employment income), 26 (median among income recipients). Metadata downloaded and checked October 20, 2026. https://www150.statcan.gc.ca/t1/wds/rest/getCubeMetadata